COLM 2026 Review Status

Paper: *The Landscape of Memorization in LLMs: Mechanisms, Measurement, and Mitigation*
Submission #3470 · Venue: COLM 2026 → EMNLP 2026 (resubmission)
Status report: 24 May 2026

Reviewer g63y (Rating: 5 / 10)

Reviewer Comment	Status in Current EMNLP Paper
Empirical study limited to HUBBLE; unclear how τ=0.08 generalizes to real models (LLaMA, GPT, etc.)	**■ Pythia-6.9b-deduped external validation added (App I.13: Loss τ_b=+0.070, ZLib τ_b=+0.084, $p<0.05$). Mistral-7B base vs. aligned comparison added (App I.14: ΔEM=+0.0008, RLHF does not suppress extraction).**
Defense component underdeveloped: paper promises 'Defense' in framework but empirical section focuses on Metric→Attack	**■ App I.17: direct unlearning experiment on Pythia-6.9b-deduped (N=50, gradient ascent, 50 steps). $\tau_b(\Delta$Loss, ΔEM)=−0.256 (p=0.009); one counterexample (ΔEM=+0.016 despite largest ΔLoss=+8.4) shows metric-verified unlearning does not certify extraction protection.**
Missing Feldman 2020 ('Does learning require memorization?') — theoretical grounding for why memorization arises	**■ Added §2.1: Feldman (2020) cited as primary theoretical grounding; rare long-tail examples may require verbatim memorization for generalization.**
Missing Ippolito et al. 2023 (verbatim vs approximate memorization distinctions)	**■ Added §2.2: Ippolito et al. (2023) cited for verbatim–approximate distinction; extraction-based metrics span strict (exact match) to lenient (ROUGE-L).**
FMARD is largely a taxonomy dressed as a causal framework; causal edges not formally validated	**■ 'Causal links' replaced with 'directed dependency links' throughout abstract, introduction, and §2. Explicit disclaimer added: links are directional/conceptual, organizing the natural pipeline flow, not interventionally causal mechanisms.**

Reviewer onNX (Rating: 5 / 10)

Reviewer Comment	Status in Current EMNLP Paper
R1 — FMARD has fuzzy boundaries between Metrics, Attacks, and Risks. Attacks seem redundant; they differ from Metrics only in intent and from Risk only in quantifiable evaluations. Can Attacks be rolled into Risk?	**■ §2 component-boundary paragraph added: same scoring function is a Metric if used offline, an Attack if deployed to classify membership — assignment depends on evaluation intent and adversary capability. Table 2 (attack taxonomy) shows Attacks differ by access requirements and leakage type; each Attack maps to different downstream Risks (many-to-many). This operationally distinguishes Attack from Risk.**
R2 — Paper hints that sequence length confounds M→A correlation but does not actually validate this claim	**■ App I.12: length-quartile stratification shows τ_b=+0.148 for shortest sequences (Q1, ≤132 tokens) vs. near-zero for Q2. App I.10 (anticorrelation analysis) shows Q4 (longest, highest-loss sequences) drives the d=256 inversion — precisely the length confound. App I.16 synthesizes Carlini 2021 and Biderman 2023 length/capacity findings.**

Reviewer Comment	Status in Current EMNLP Paper
Q1 — Line 91: 'Sampling and decoding' is an attacker-side intervention, not a Factor affecting what gets memorized	**■ Renamed to 'Inference-time decoding' (Factor 4). §2.1 now explains it is adversary-accessible at test time, not a fixed training property; decoding is de facto part of Attack configuration.**
Q2 — Table 1: Effects of 'Training duration' and 'Alignment' not discussed in §2.1	**■ Factors 6 (Training duration) and 7 (Alignment/RLHF) added to §2.1 prose and to Table 1 with effect directions and citations.**
Q3 — Line 156: Overlapping definitions of Attack and Metric — unclear why Metrics would be identical if one Attack succeeds while another fails	**■ §2 boundary paragraph clarifies: Metrics are offline statistical scores; Attacks are adversarial procedures requiring resources and model access. The same function (e.g., loss threshold) is a Metric offline and part of an Attack when deployed for membership classification.**
Q4 — Table 3: Column heading is 'FMARD Link' but most entries are not F/M/A/R/D notation	**■ Table 3 'FMARD Link Targeted' column now uses arrow notation throughout: F (Factor), F→M (Training), M→A (Post-training), A→R (Inference).**
Q5 — Line 184: No citation for 'strict privacy budgets degrade utility'	**■ Added \citep{li2021large, abadi2016deep} at the relevant sentence.**
Q6 — Line 212: 'Compressing the signal' is unclear jargon	**■ Replaced with 'shrinking the loss gap between members and non-members (the signal MIA relies on) and reducing its discriminative power'.**
Q7 — Line 260: 'Target extraction performance' is undefined	**■ Parenthetical added: '(defined as zero greedy exact match on the forget set, the standard success criterion for unlearning)'.**
Q8 — §4.5: PII attack success depending on adversary capability (background knowledge) already established in HUBBLE paper	**■ Added paragraph distinguishing new finding: per-field $\|\tau_b\|<0.12$ between document-level Metrics and field-level extraction outcomes is new — no forward-pass metric predicts which specific PII fields are extractable. This is a structural M→A diagnostic gap not established by HUBBLE.**
Suggestion 1 — Line 51: Not enough setup to understand what 'AUC gap' means	**■ 'AUC gap' terminology no longer appears unexplained in the introduction. Early intro uses 'metric–risk gap'. ΔAUCt appears in contribution bullets with context 'vs. an unexposed model'.**

Reviewer mCRV (Rating: 5 / 10)

Reviewer Comment	Status in Current EMNLP Paper
R1 — FMARD repeatedly presented as causal with directed dependencies, but relationships are not formally causal in interventionist sense; empirical study demonstrates correlations, not mechanisms	**■ 'Causal links' → 'directed dependency links' throughout. §2 operational definition revised: Condition (1) changed from 'necessary precondition' (empirically false at d=0) to modulation-based formulation. Explicit disclaimer: links are directional/conceptual, not interventionally causal.**

Reviewer Comment	Status in Current EMNLP Paper
R2 — Experiments limited to single model family, base models only, exclude white-box attacks (soft-prompt optimization); gap between framework breadth and empirical validation is substantial	**■ App I.13: Pythia-6.9b-deduped external validation (600 sequences, heterogeneous natural duplication). App I.14: Mistral-7B base vs. instruction-tuned (aligned model). App I.15: White-box attacks addressed analytically — τ_b=0.08 greedy is conservative lower bound; gradient-based adversary raises EM for sequences already flagged by metrics.**
R3 — Defense component mostly discussed through literature synthesis rather than directly evaluated	**■ App I.17: Direct unlearning experiment (Pythia-6.9b-deduped, N=50 top-EM sequences, gradient ascent on last 2 layers). Shows metric-verified unlearning (ΔLoss>0 in 50/50) does not certify extraction protection (ΔEM<0 in only 49/50; one sequence ΔEM=+0.016 despite largest ΔLoss=+8.4).**
Q — Concrete example where FMARD leads to different practical recommendation than without framework	**■ §3 worked example added: without FMARD, practitioner auditing copyright risk selects perplexity (cheap, widely-used). FMARD reveals τ_b≈0.035 on literary text — near-zero predictive signal. FMARD-guided recommendation is qualitatively different: use extraction attacks directly, stratified by content type. Without M→A decomposition, practitioner has no principled basis for knowing low perplexity says nothing about extractability.**
Q — What makes FMARD specifically causal rather than structured taxonomy with directed dependencies?	**■ Paper no longer claims formal causality — 'directed dependency links' framing adopted. FMARD is positioned as a structured taxonomy with directional conceptual dependencies reflecting the natural temporal/logical order of the memorization pipeline.**
Q — How sensitive are results to choice of extraction attack and decoding strategy?	**■ App I.12 (confounder analysis): greedy vs. sampling EM compared across 16 metric–attack pairs. Loss vs. Sampling EM swings to τ_b≈–0.55 at d=256, while Loss vs. Greedy EM stays at –0.52. Best pair (MinK%++ vs. Static Greedy EM) reaches τ_b=0.292; all sampling-based pairs lower.**
Q — Are seven metrics equally appropriate across all three domains, or should different domains require different metric choices?	**■ Domain-sensitive metric note added §4.3: metric choice is domain-sensitive. App I.13 shows Min-K% and MinK%++ are non-significant and negative on natural low-duplication text (Pythia/WikiText-103), while Loss and ZLib remain positive — calibrated count-based metrics invert in natural settings and should not be primary metrics without domain validation.**

Reviewer AUwY (Rating: 4 / 10)

Reviewer Comment	Status in Current EMNLP Paper
R1 — 'Causal' framing not empirically justified. τ=0.08 establishes that two measurement families are not interchangeable (correlation finding), but does not establish directionality, mechanism, or counterfactual dependence between FMARD components	**■ 'Causal links' → 'directed dependency links' throughout. §2 operational definition: directed dependency link X→Y asserts that properties of X modulate the strength or character of Y (modulation-based, not necessity-based). Explicit disclaimer in §2: links are directional/conceptual, organizing the memorization pipeline by natural flow. No counterfactual causality claimed.**

Reviewer Comment	Status in Current EMNLP Paper
R2 — Controlled study evaluates M→A link exclusively. Remaining links (F→M, A→R, Defense→Factor, cross-link interference) supported only by citations to existing literature, not new controlled experiments	■ **Explicitly acknowledged in Limitations: only M→A directly tested. Appendix adds partial evidence: I.13 (M→A on natural text), I.14 (Defense→A: RLHF as A-output Defense), I.17 (Defense→M and M→A gap). Contributions section (3) explicitly scopes to M→A on HUBBLE; F→M, A→R, Defense links flagged as synthesis-supported open empirical questions.**
R3 — Three failure modes (link mismatch, factor confounding, cross-link interference) are recognizable instances of well-understood methodological problems that exist independently of FMARD	■■ **Partially addressed. §3 worked example shows FMARD changes a practitioner recommendation (concrete practical value added). However, the philosophical point — that these failure modes are recognizable without FMARD — is inherent to any systematization-of-knowledge contribution. FMARD's value is in naming, systematizing, and providing a common vocabulary; this is acknowledged but the underlying critique cannot be fully resolved through text changes.**
R4 — Threat-model-driven guidelines (§5) are useful but do not obviously require FMARD to derive. Copyright auditing recommendation follows directly from observing loss is uncorrelated with extractability	■ **§3 worked example directly addresses this: without FMARD's M→A decomposition, a practitioner has no principled basis for knowing that low perplexity on literary text says nothing about extractability. FMARD provides the structural framework that makes τ_b≈0.035 interpretable as a link-specific finding rather than a domain-specific anomaly.**

Overall Assessment

All substantive reviewer concerns are addressed in the current EMNLP 2026 submission except **AUwY R3** (failure modes are well-understood methodological problems that predate FMARD). This is a philosophical critique inherent to systematization-of-knowledge contributions and is appropriately handled by acknowledging scope rather than claiming FMARD discovered the modes. The paper is well-positioned for EMNLP acceptance after stripping the changes-package markup prior to final submission.

■ **Fully addressed in current paper**	■■ **Partially addressed; residual philosophical concern**